INT374: DATA ANALYTICS WITH POWER BI

PROJECT REPORT

(Project Semester August-January 2025)

Analysis of Historical Rainfall Patterns in India

Discipline of CSE
Computer Science & Engineering
Lovely Professional University, Phagwara

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CERTIFICATE

This is to certify that charesma kotapati bearing Registration No. 12302465 has successfully completed the course project titled **"Analysis of Historical Rainfall Patterns in India"** under my guidance and supervision for the course INT374. The work presented is original and meets the academic requirements.

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DECLARATION

I, Charesma Kotapati, student of B.Tech Computer Science & Engineering under CSE Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine. This project work is original and has not been submitted elsewhere for any academic award.

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TABLE OF CONTENTS

S.No	Title	Page No.
1	Title Page	1
2	Certificate	2
3	Declaration	3
4	Acknowledgement	4
5	Table of Contents	5
6	List of Figures	6
7	Chapter 1 – Introduction	7-8
8	Chapter 2 – Problem Statement	9
9	Chapter 3 – Objectives	10-12
10	Chapter 4 – Source of Dataset	13
11	Chapter 5 – Methodology	14-15
12	Chapter 6 – Tools and Technologies Used	16
13	Chapter 7 – Data collection and preparation	16-19
14	Chapter 8 – Dashboard Design & Visualizations	20-27
15	Chapter 9 – Insights and Results	28
16	Chapter 10 – Conclusion	29
17	Chapter 11 – References	30

LIST OF FIGURES

FIGURE NO.	NAME OF THE FIGURE	PAGE NO
FIGURE 1	Average Annual Rainfall Trend in India (1901–2017)	20
FIGURE 2	Average Rainfall by Season	20
FIGURE 3	Rainfall Intensity Distribution (Low, Moderate, High)	21
FIGURE 4	Top 10 Sub-divisions by Average Rainfall	21
FIGURE 5	Bottom 10 Sub-divisions by Average Rainfall	22
FIGURE 6	Spatial Distribution of Average Rainfall Across India	22
FIGURE 7	Rainfall Contribution – Coastal vs Inland Regions	23
FIGURE 8	Extreme Rainfall Events (Top 15 Years)	23
FIGURE 9	Year-wise Rainfall Variability (Extreme Perspective)	24
FIGURE 10	High-Risk Regions – Average vs Extreme Rainfall	24
FIGURE 11	Seasonal Rainfall Contribution Over Time	25
FIGURE 12	Extreme Rainfall Events by Decade	26
FIGURE 13	Monthly Rainfall Intensity Across Years (Heatmap Matrix)	27

CHAPTER 1

INTRODUCTION

Rainfall is one of the most critical climatic factors influencing India's economy, agriculture, water resources, and environmental sustainability. As a country with diverse geographical regions—ranging from coastal belts and plains to plateaus and mountainous terrains—India experiences significant variation in rainfall across different regions and seasons. The Indian monsoon system, particularly the southwest monsoon, plays a dominant role in determining annual rainfall distribution and directly impacts food security, groundwater recharge, and ecosystem balance.

Over the years, irregular rainfall patterns, changing seasonal behaviour, and the increasing frequency of extreme rainfall events have raised serious concerns among policymakers, environmentalists, and researchers. Floods, droughts, and water scarcity have become recurring challenges in many parts of the country, highlighting the need for detailed rainfall analysis rather than relying solely on annual averages.

Traditional rainfall analysis methods, such as static reports or tabular summaries, often fail to capture deeper insights like rainfall variability, regional disparities, seasonal dominance, and long-term extreme patterns. With the availability of long-term historical climate data and advancements in data analytics tools, it has become possible to explore rainfall behaviour in a more interactive, visual, and insightful manner.

Microsoft Power BI provides a powerful platform for transforming large and complex datasets into meaningful dashboards through data modelling, advanced calculations, and interactive visualizations. By integrating data preparation, analytical measures, and visualization techniques, Power BI enables users to identify hidden trends, patterns, and anomalies effectively.

This project focuses on the **analysis of historical rainfall data in India from 1901 to 2017**, covering multiple sub-divisions, seasons, and time periods. The study aims to examine long-term rainfall trends, seasonal contributions, spatial distribution, variability across years, and

clustering of extreme rainfall events. Special emphasis is placed on identifying high-risk regions that experience both high average rainfall and extreme fluctuations.

Through a set of well-designed interactive dashboards, this project demonstrates how visual analytics can support climate-related decision-making and improve understanding of complex rainfall patterns. The insights derived from this analysis can be useful for academic study, environmental research, water resource planning, and disaster risk assessment.

CHAPTER 2

PROBLEM STATEMENT

India's rainfall pattern is highly complex and varies significantly across regions, seasons, and years due to its diverse geography and dependence on the monsoon system. While certain regions receive abundant rainfall during specific seasons, others face irregular rainfall distribution, prolonged dry spells, or sudden extreme rainfall events. This uneven distribution poses serious challenges to agriculture, water resource management, disaster preparedness, and long-term environmental sustainability.

One of the major challenges in rainfall analysis is that conventional approaches often rely on annual or seasonal averages, which mask critical variations such as inter-annual volatility, regional disparity, and the occurrence of extreme rainfall events. Such aggregated views fail to capture important insights like which regions are consistently vulnerable to floods, which areas experience high variability despite high average rainfall, and whether extreme rainfall events are becoming more concentrated over time.

Additionally, rainfall data is typically available in large, multi-year datasets that are difficult to interpret using static tables or simple charts. Without proper data preparation, modeling, and visualization, valuable information remains hidden within the dataset, limiting its practical usefulness for analysis and decision-making.

There is a need for an interactive and analytical approach that can transform historical rainfall data into meaningful insights by enabling users to explore patterns across multiple dimensions such as time, season, and geography. An effective solution should allow dynamic filtering, comparison of regions, identification of high-risk areas, and analysis of long-term trends and variability.

This project addresses these challenges by leveraging Microsoft Power BI to analyse historical rainfall data for India from 1901 to 2017. By applying data cleaning, transformation, analytical measures, and interactive visual dashboards, the project aims to provide a comprehensive understanding of rainfall behaviour. The focus is not only on average rainfall trends but also on rainfall variability, seasonal dominance, regional extremes, and decade-wise clustering of extreme events, thereby offering a deeper and more practical insight into India's rainfall patterns.

CHAPTER 3

OBJECTIVES OF THE PROJECT

The main objective of this project is to analyse historical rainfall data of India and present key climatic insights through interactive dashboards developed using Microsoft Power BI. The project focuses on understanding rainfall trends, seasonal behaviour, regional variability, and extreme rainfall patterns to support data-driven environmental and climatic analysis. The specific objectives of the project, based on the implemented dashboards, are as follows:

i. To analyse overall rainfall trends using key rainfall indicators.

This objective aims to provide a high-level understanding of India's rainfall behaviour by analysing average rainfall, maximum rainfall, minimum rainfall, and rainfall variability over time. KPI cards and trend-based visualizations are used in the Overview dashboard to summarize long-term rainfall patterns.

ii. To examine seasonal rainfall contribution across different time periods.

This objective focuses on analysing rainfall distribution across meteorological seasons such as JF, MAM, JJAS, and OND. Column charts are used to identify dominant seasons and understand the role of monsoon rainfall in shaping annual totals.

iii. To classify rainfall observations based on intensity levels.

This objective aims to categorize rainfall into Low, Moderate, and High intensity classes. A donut chart is used to visualize the proportion of rainfall observations under each category, helping assess rainfall severity and potential impact.

iv. To identify sub-divisions with the highest and lowest rainfall levels.

This objective focuses on comparing rainfall across Indian sub-divisions to identify regions receiving extreme wet or dry conditions. Bar charts and tables are used to highlight top and bottom rainfall-receiving regions.

v. To visualize the spatial distribution of rainfall across India.

This objective aims to represent geographical rainfall patterns using map visualizations. The spatial analysis helps in identifying rainfall-rich zones and regions vulnerable to low rainfall or extreme events.

vi. To compare rainfall contribution between coastal and inland regions.

This objective focuses on understanding how geographical location influences rainfall consistency and intensity. Donut charts are used to compare rainfall contribution between coastal and inland sub-divisions.

vii. To analyse extreme rainfall events and their occurrence over time.

This objective aims to identify years with unusually high rainfall values and analyse extreme rainfall behaviour. Line and column charts are used to highlight abnormal rainfall events and their temporal distribution.

viii. To study rainfall variability across years.

This objective focuses on analysing inter-annual rainfall variability rather than relying solely on average values. Variability indicators and extreme rainfall measures are used to assess fluctuations across different years.

ix. To identify high-risk regions based on rainfall behaviour.

This objective aims to detect sub-divisions that experience both high average rainfall and extreme rainfall events. A scatter plot is used to identify risk-prone regions vulnerable to floods and rainfall-related hazards.

x. To analyse rainfall behaviour across decades.

This objective focuses on understanding long-term rainfall patterns by aggregating data decade-wise. Column charts are used to identify clustering of extreme rainfall events across different decades.

xi. To examine seasonal rainfall contribution over time.

This objective aims to analyse how the contribution of different seasons to total rainfall has changed over the years. Stacked column charts are used to study shifts in seasonal dominance.

xii. To analyse monthly rainfall variability across years.

This objective focuses on identifying months that contribute most to rainfall intensity and variability. A matrix heatmap is used to visualize month-wise rainfall patterns across multiple years.

xiii. To enable interactive and flexible rainfall analysis using slicers and filters.

This objective aims to enhance user interactivity by allowing filtering of data based on year, sub-division, season, and month. Slicers are implemented across dashboards to support dynamic exploration and customized analysis.

CHAPTER 4

SOURCE OF THE DATASET

The dataset used in this project has been obtained from **data.gov.in**, which is the official open data platform of the Government of India. Data.gov.in provides publicly accessible, authoritative datasets published by various government departments and organizations, supporting transparency, research, and data-driven analysis. The platform is widely used by students, researchers, analysts, and policymakers for academic and analytical purposes.

The specific dataset used for this project contains **historical rainfall data of India** and includes records spanning multiple years across various meteorological sub-divisions. (https://www.data.gov.in/resource/sub-divisional-monthly-rainfall-1901-2017)

This dataset represents real-world rainfall observations collected and maintained by official government sources, making it highly reliable for studying rainfall trends, seasonal behaviour, regional variability, and extreme rainfall events. The structured nature of the dataset allows effective analysis using business intelligence and data visualization tools such as **Microsoft Power BI**.

The data provided through data.gov.in is well-documented and intended strictly for **educational, research, and analytical purposes**. It does not contain any sensitive or personally identifiable information. The open nature of the dataset ensures ethical usage, reproducibility of analysis, and transparency in reporting results.

Overall, the rainfall dataset sourced from data.gov.in serves as a credible and appropriate data source for this project. It enables detailed analysis of India's rainfall patterns and supports the development of interactive Power BI dashboards for understanding long-term trends, seasonal variations, regional disparities, and rainfall variability

CHAPTER 5

METHODOLOGY

The methodology adopted in this project focuses on transforming historical rainfall data into meaningful insights using Microsoft Power BI. A structured workflow involving data preparation, modelling, analytical computation, and visualization was followed to meet the project objectives.

4.1 Project Workflow

1. Dataset Identification and Import

- The historical rainfall dataset was obtained from **data.gov.in**, the official open data platform of the Government of India.
- The dataset includes rainfall records across multiple years and Indian meteorological sub-divisions.
- The data was imported into **Microsoft Power BI** using standard file import connectors.

2. Data Cleaning and Preparation

- Initial data exploration and cleaning were performed using **Power Query Editor**.
- Data types were standardized and missing or inconsistent rainfall values were examined and corrected.
- Additional columns such as rainfall category, decade, and month were created to support advanced analysis.

3. Data Modelling

- A **fact–dimension model** was designed to organize rainfall data efficiently.
- Rainfall values were stored in a central fact table, supported by dimension tables for year, sub-division, season, and month.
- Relationships were created to enable correct filtering and interaction across dashboards.

4. DAX Measures Creation

- Custom **DAX measures** were created to calculate average, maximum, and minimum rainfall.

- Rainfall variability and extreme rainfall indicators were computed using analytical expressions.
- These measures supported KPI cards and comparative analysis across dashboards.

5. Dashboard Visualization Design

- Interactive dashboards were developed using charts, maps, KPI cards, and matrices in Power BI.
- Separate dashboards were designed for overview analysis, regional analysis, and rainfall variability patterns.
- Slicers and filters were added to enable dynamic analysis by year, region, season, and month.

6. Result Interpretation and Insight Derivation

- Insights were derived by examining trends, variability, and extreme rainfall patterns.
- Comparative analysis helped identify high-risk regions and dominant rainfall seasons.
- Interactive visuals allowed flexible exploration and deeper understanding of rainfall behaviour.

7. Documentation and Reporting

- The entire project workflow, analysis, and findings were documented in a structured report.
- Dashboard visuals were explained clearly to support academic evaluation and presentation.
- The report was prepared in an ethical and reproducible manner suitable for submission.

CHAPTER 6

TOOLS AND TECHNOLOGIES USED

The development of this project involved various tools, software platforms, and analytical technologies. These tools were selected based on their ability to support data preparation, modelling, analysis, and visualization of historical rainfall data in an effective and interactive manner.

6.1 Microsoft Power BI Desktop

Microsoft Power BI Desktop was the primary tool used for this project. It was utilized for:
• importing the rainfall dataset,
• performing data modelling using fact and dimension tables,
• creating DAX measures and KPI calculations,
• designing interactive dashboards and visualizations, and
• enabling slicers and filters for dynamic rainfall analysis.

6.2 Power Query Editor

Power Query Editor was used for data preprocessing and transformation tasks such as:
• cleaning and validating rainfall values,
• handling missing or inconsistent data,
• creating derived columns such as rainfall category and decade, and
• reshaping monthly rainfall data using unpivoting techniques.

6.3 DAX (Data Analysis Expressions)

DAX was used to create analytical measures and calculations required for rainfall analysis. It was applied for:
• computing average, maximum, and minimum rainfall values,
• calculating rainfall variability and extreme rainfall indicators, and
• generating context-aware calculations for KPI cards and visual comparisons.

6.4 CSV Dataset (data.gov.in)

The raw rainfall dataset was available in **CSV format** and sourced from **data.gov.in**. The dataset was initially reviewed for structure and completeness before being loaded into Power BI, ensuring accurate interpretation and effective analysis of historical rainfall patterns.

CHAPTER 7

DATA COLLECTION & PREPARATION

Data collection and preparation is a critical stage in any data analytics project, as the quality of Data collection and preparation is a critical stage in any data analytics project, as the quality of analysis and insights largely depends on the quality of the input data. In this project, careful attention was given to collecting an authentic rainfall dataset and preparing it systematically to ensure accuracy, consistency, and suitability for analysis using Microsoft Power BI.

7.1 Data Collection

The dataset used for this project was collected from **data.gov.in**, the official open data platform of the Government of India, which provides reliable and publicly accessible datasets for research and educational purposes. The selected dataset contains **historical rainfall data of India**, representing rainfall measurements recorded across different meteorological sub-divisions.

The dataset includes structured rainfall information such as:
• Sub-division name
• Year
• Monthly rainfall values
• Seasonal rainfall (JF, MAM, JJAS, OND)
• Annual rainfall totals

The dataset was downloaded in **CSV format** and imported into **Microsoft Power BI** using the built-in Text/CSV data connector. The data represents real-world meteorological observations and is suitable for rainfall trend analysis, seasonal study, and regional comparison.

7.2 Data Understanding

After importing the dataset into Power BI, an initial data exploration was conducted to understand the structure, columns, and data types. This step helped in identifying:
• Key fields such as year, sub-division, month-wise rainfall, seasonal rainfall, and annual rainfall
• The temporal range of the dataset spanning multiple decades
• The numerical nature of rainfall measurements and categorical nature of sub-divisions
• Presence of zero or missing rainfall values in certain months

Understanding the dataset at this stage ensured appropriate preprocessing and modelling decisions before performing analytical tasks.

7.3 Data Cleaning

Data cleaning was carried out using **Power Query Editor** in Power BI. The following steps were performed to improve data quality:
• Verification and correction of data types for rainfall values and year fields
• Handling of missing, zero, or inconsistent rainfall values where necessary
• Removal of redundant columns that were not required for analysis
• Standardization of sub-division names and categorical fields
• Validation of numerical rainfall values to ensure accuracy and consistency

These steps ensured that the dataset was clean, reliable, and suitable for further analysis.

7.4 Data Transformation

To support time-based and advanced rainfall analysis, additional transformations were applied to the dataset:
• Creation of a **Decade** column to enable long-term trend and decadal analysis
• Transformation of monthly rainfall columns using **unpivoting** to allow month-wise analysis
• Creation of **Rainfall Category** (Low, Moderate, High) to support intensity-based analysis

These transformations enhanced analytical flexibility and enabled multiple perspectives on rainfall behaviour.

7.5 Data Integration and Modelling Preparation

The cleaned and transformed dataset was prepared for analytical modelling by:
• Treating rainfall measurements as the central **fact table**
• Creating supporting **dimension tables** for Year, Sub-division, Season, and Month
• Ensuring consistent key fields to enable correct relationships and filter propagation

This preparation ensured smooth integration of data into the data model and enabled interactive cross-filtering across dashboards.

7.6 Prepared Dataset for Analysis

After completing the data collection and preparation steps, the dataset was fully ready for analysis and visualization in Power BI. The prepared data supported:
• KPI calculations using DAX measures
• Long-term rainfall trend and variability analysis
• Seasonal and monthly rainfall pattern analysis

• Regional and geographical rainfall comparison
• Interactive dashboard filtering using slicers

Overall, the data collection and preparation process ensured that the rainfall dataset was accurate, consistent, and well-structured, forming a strong foundation for the analytical dashboards developed in this project.

CHAPTER 8

DASHBOARD DESIGN & VISUALIZATIONS

This project uses multiple interactive dashboards designed in **Microsoft Power BI** to analyze historical rainfall patterns in India. Each dashboard focuses on a specific analytical perspective such as long-term trends, seasonal behaviour, regional variability, and extreme rainfall events. The dashboards make use of KPI cards, charts, maps, matrices, and slicers to effectively communicate rainfall insights.

8.1 Overview Rainfall Dashboard

The Overview Rainfall dashboard provides a high-level summary of rainfall patterns across India and helps users understand overall rainfall behaviour at a national level.

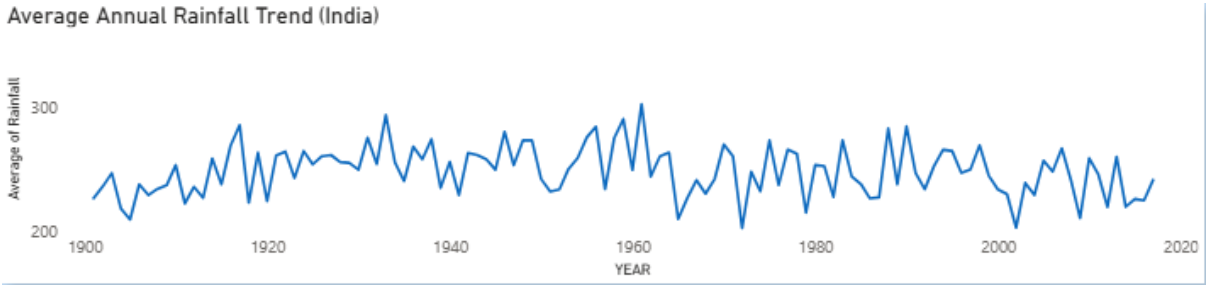
1. Overall Rainfall Trend Analysis

• **Objective:**
To analyse long-term trends in average annual rainfall across India over multiple years.

• **Visualization Used:**
o Line Chart

• **Analysis:**
The line chart displays the variation in average annual rainfall over time, helping identify long-term fluctuations, stability, or abnormal changes in rainfall behaviour.

FIG 1. Average Annual Rainfall Trend in India



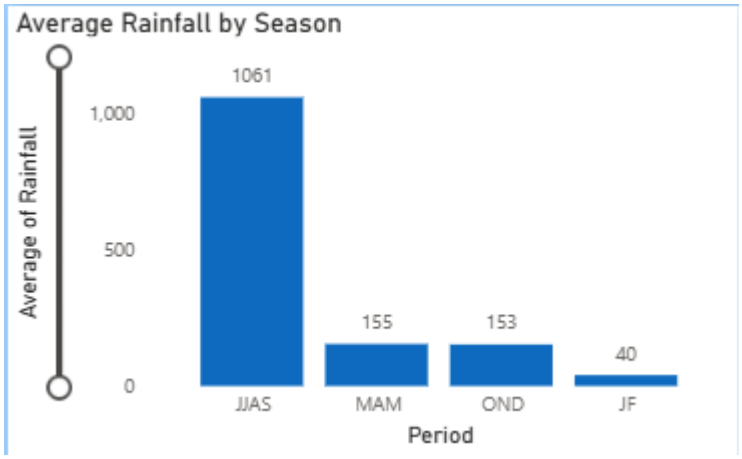
2. Seasonal Rainfall Contribution Analysis

• **Objective:**
To examine the contribution of different seasons (JF, MAM, JJAS, OND) to total annual rainfall.

• **Visualization Used:**
o Column Chart

• **Analysis:**
The column chart highlights the dominance of monsoon rainfall, particularly the JJAS season, and helps compare seasonal rainfall contribution clearly.

FIG 2. Average Rainfall by Season



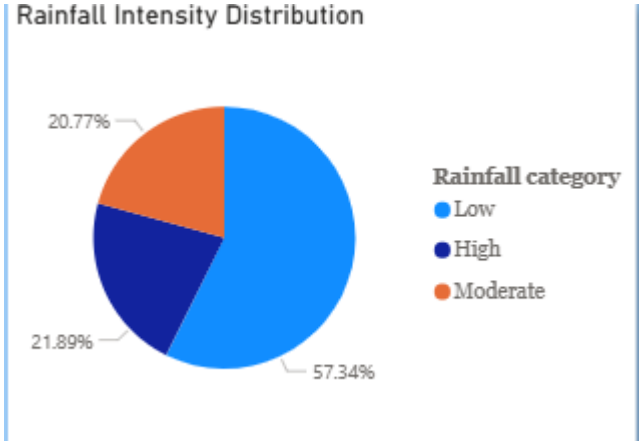
3. Rainfall Intensity Distribution

• **Objective:**
To classify rainfall observations into Low, Moderate, and High intensity categories.

• **Visualization Used:**
o Donut Chart

• **Analysis:**
The donut chart shows the proportion of rainfall observations under each intensity category, helping assess the severity and distribution of rainfall levels.

FIG 3. Rainfall Intensity Distribution



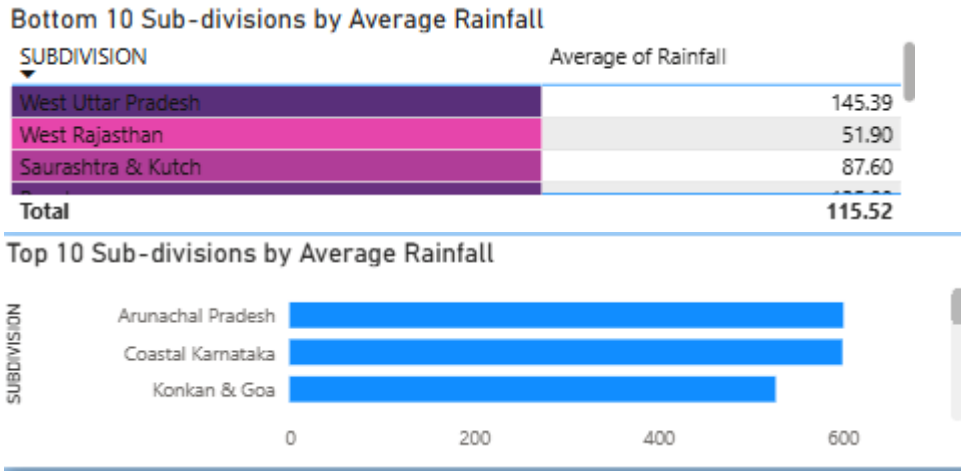
4. Sub-Division Level Rainfall Comparison

• **Objective:**
To identify sub-divisions receiving the highest and lowest average rainfall.

• **Visualization Used:**
o Bar Chart and Table

• **Analysis:**
The visual comparison highlights regional disparities by displaying top and bottom rainfall-receiving sub-divisions, helping identify extremely wet and dry regions.

FIG 4. Top Rainfall-Receiving Sub-Divisions
FIG 5. Lowest Rainfall-Receiving Sub-Divisions



8.2 Regional Rainfall Analysis Dashboard

The Regional Rainfall Analysis dashboard focuses on spatial and geographical patterns of rainfall across India.

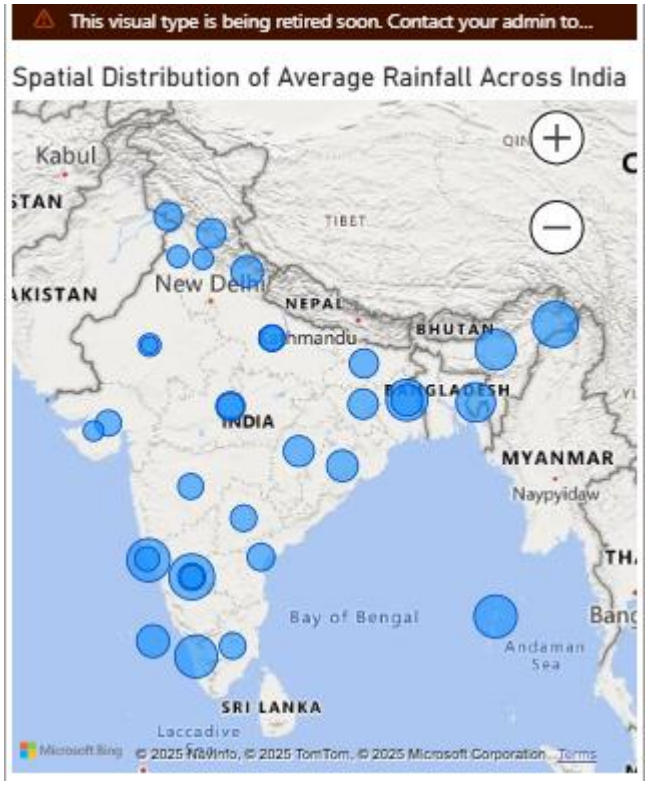
5. Geographical Distribution of Rainfall

• **Objective:**
To visualize the spatial distribution of rainfall across Indian sub-divisions.

• **Visualization Used:**
o Map (Bubble Map)

• **Analysis:**
The map visualization highlights rainfall concentration across regions, enabling identification of rainfall-rich zones and regions prone to low or extreme rainfall.

FIG 6. Spatial Distribution of Rainfall Across India



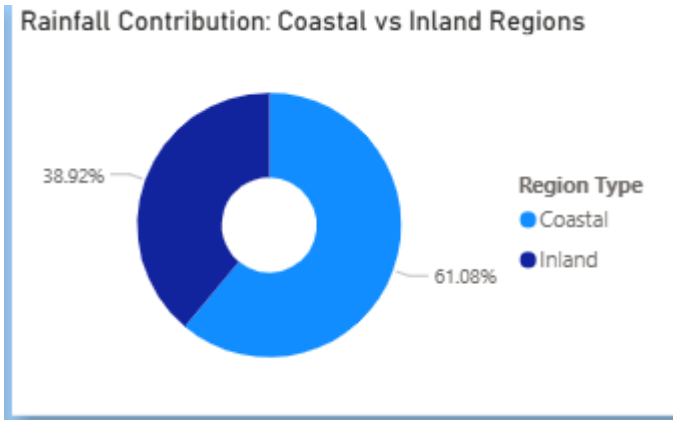
6. Coastal vs Inland Rainfall Comparison

• **Objective:**
To compare rainfall contribution between coastal and inland regions.

• **Visualization Used:**
o Donut Chart

• **Analysis:**
The donut chart compares rainfall contribution between coastal and inland sub-divisions, highlighting the influence of geographical location on rainfall consistency.

FIG 7. Coastal vs Inland Rainfall Contribution



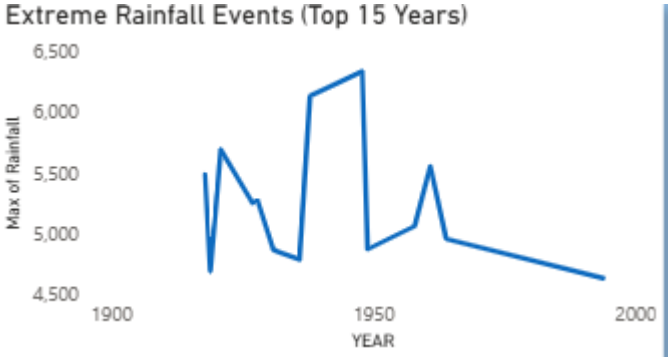
7. Extreme Rainfall Events Analysis

• **Objective:**
To identify years with extreme rainfall events.

• **Visualization Used:**
o Column Chart

• **Analysis:**
The chart highlights years with unusually high rainfall values, helping identify abnormal rainfall events and periods of increased climatic risk.

FIG 8. Extreme Rainfall Events by Year



8.3 Rainfall Variability & Pattern Analysis Dashboard

This dashboard focuses on advanced analytical insights related to rainfall variability, risk-prone regions, and long-term rainfall behaviour.

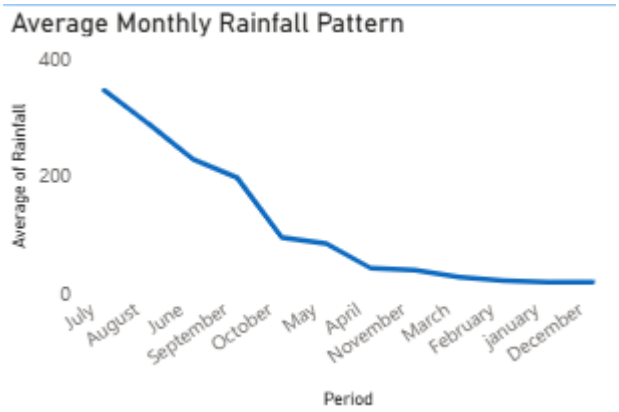
8. Rainfall Variability Across Years

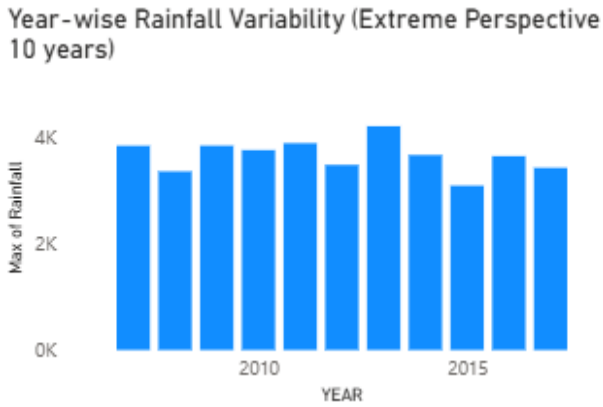
• **Objective:**
To analyse year-wise rainfall variability using extreme rainfall indicators.

• **Visualization Used:**
o Column Chart

• **Analysis:**
The visualization highlights variations in rainfall intensity across years, focusing on volatility rather than average rainfall values.

FIG 9. Year-wise Rainfall Variability





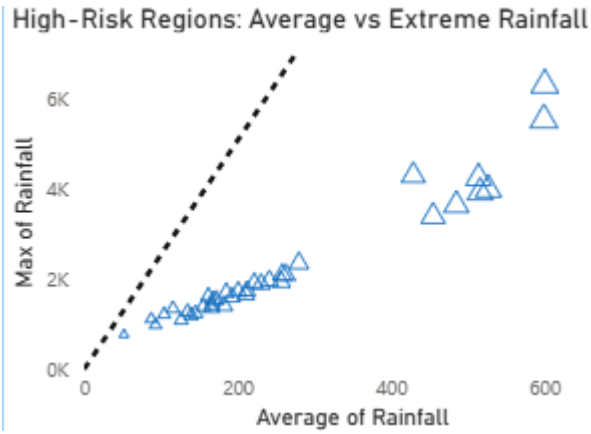
9. Identification of High-Risk Regions

• **Objective:**
To identify sub-divisions experiencing both high average rainfall and extreme rainfall events.

• **Visualization Used:**
o Scatter Plot

• **Analysis:**
The scatter plot compares average rainfall against extreme rainfall values, helping identify risk-prone regions vulnerable to floods and rainfall-related hazards.

FIG 10. High-Risk Regions – Average vs Extreme Rainfall



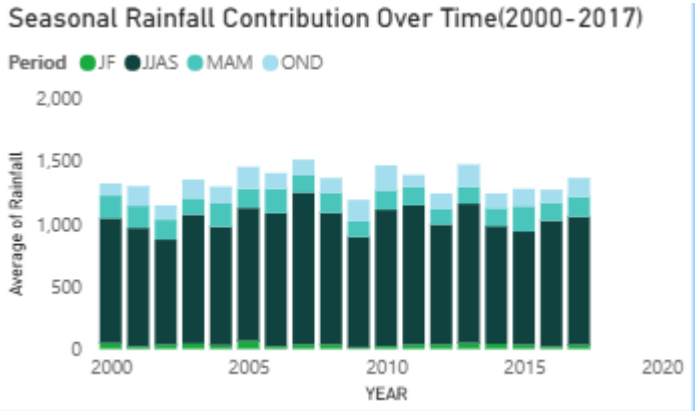
10. Seasonal Rainfall Contribution Over Time

• **Objective:**
To study how seasonal rainfall contribution has changed over time.

• **Visualization Used:**
o Stacked Column Chart

• **Analysis:**
The stacked column chart shows the contribution of different seasons across years, helping identify shifts in seasonal dominance and rainfall behavior.

FIG 11. Seasonal Rainfall Contribution Over Time



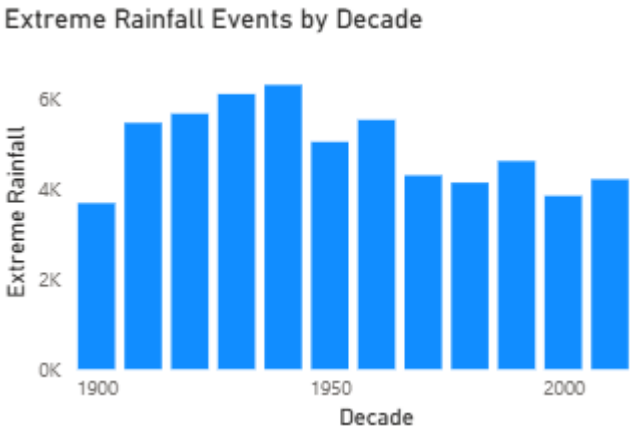
11. Extreme Rainfall Events by Decade

• **Objective:**
To analyse extreme rainfall behaviour across decades.

• **Visualization Used:**
o Column Chart

• **Analysis:**
The decade-wise analysis helps identify clustering of extreme rainfall events over long-term periods.

FIG 12. Extreme Rainfall Events by Decade



12. Monthly Rainfall Variability Analysis

• **Objective:**
To identify months contributing most to rainfall variability.

• **Visualization Used:**
o Matrix (Heatmap)

• **Analysis:**
The heatmap-style matrix displays month-wise rainfall intensity across years, highlighting months with higher rainfall concentration and variability.

FIG 13. Monthly Rainfall Variability Across Years

Monthly Rainfall Intensity Across Years

Period	1901	1902	1903	1904	1905
ANNUAL	1,284.21	1,337.30	1,399.91	1,236.77	1,18
April	46.39	45.77	16.61	41.40	3
August	301.33	239.80	302.35	217.01	24
December	10.81	33.15	22.31	14.77	1
February	40.51	8.31	11.97	9.39	2
january	35.22	8.52	15.48	11.70	2
JF	75.72	16.81	27.45	21.08	4
JJAS	921.31	1,018.39	1,052.60	933.87	87
July	291.21	351.15	365.06	311.97	28
June	194.61	168.82	170.80	237.49	16

CHAPTER 9

INSIGHTS AND RESULTS

The Power BI dashboards developed in this project provide clear and meaningful insights into historical rainfall patterns across India. The Overview dashboard highlights long-term rainfall behaviour using key rainfall indicators such as average rainfall, maximum rainfall, and rainfall variability, enabling a comprehensive understanding of overall rainfall trends.

Seasonal analysis reveals that rainfall contribution is highly dependent on specific seasons, with the monsoon period contributing the largest share of annual rainfall. The rainfall intensity classification shows that a significant portion of observations fall under moderate rainfall, while extreme rainfall events occur less frequently but have a strong impact. Sub-division level analysis highlights clear regional disparities, where certain regions consistently receive high rainfall while others remain rainfall deficient.

Regional and spatial analysis indicates that rainfall distribution is uneven across India, with coastal regions generally receiving more consistent rainfall compared to inland regions. Extreme rainfall analysis identifies specific years and decades with unusually high rainfall values, indicating periods of increased climatic variability. The scatter plot analysis further reveals high-risk regions that experience both high average rainfall and extreme rainfall events, making them more vulnerable to floods and rainfall-related hazards.

Overall, the results demonstrate that Power BI effectively transforms large and complex rainfall datasets into interactive and insightful visual dashboards. The combination of trend analysis, variability assessment, and spatial visualization supports better understanding of rainfall behaviour and provides a strong foundation for environmental and climatic analysis.

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CHAPTER 10

CONCLUSION

This project successfully demonstrates the application of **Microsoft Power BI** for analysing historical rainfall data of India and converting raw meteorological data into meaningful and interpretable insights. Using interactive dashboards, the project provides a comprehensive understanding of rainfall trends, seasonal behaviour, regional variability, and extreme rainfall patterns.

The Overview dashboard enables effective evaluation of long-term rainfall trends and seasonal dominance, while the Regional Analysis dashboard highlights geographical rainfall distribution and differences between coastal and inland regions. The Rainfall Variability and Pattern Analysis dashboard offers advanced insights into rainfall volatility, high-risk regions, and decade-wise extreme rainfall clustering.

Overall, the project highlights the effectiveness of Power BI as a data analytics and visualization tool for climate-related studies. The dashboards developed are interactive, easy to interpret, and suitable for both academic evaluation and practical analysis. This project demonstrates how data-driven approaches can enhance understanding of environmental patterns and support informed decision-making related to water resource management and climate analysis.

CHAPTER 11

REFERENCES

The references listed below include dataset sources, software tools, and supporting materials used to complete this project. These references support the reliability of the dataset, the analytical methodology adopted, and the tools used for visualization and analysis.

11.1 Dataset Sources

1. **Government of India – Open Data Platform (data.gov.in)**
   Historical Rainfall Data of India
   Available at: https://data.gov.in
2. **India Meteorological Department (IMD)**
   Climate and Rainfall Information
   Available at: https://mausam.imd.gov.in

11.2 Tools and Software

1. **Microsoft Power BI Desktop** – Data modelling, visualization, dashboard design, and DAX computations.
   Available at: https://powerbi.microsoft.com
2. **Power Query Editor** – Data cleaning, transformation, and preparation within Power BI.
3. **Microsoft Excel** – Preliminary dataset review and structure verification before data import.